DocsAI

Comprehensive Standard Operating Procedure (SOP)

Version 1.0

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1. Executive Summary

DocsAI is an AI-powered enterprise knowledge platform. It ingests business documents, extracts structured knowledge, creates semantic embeddings, and enables intelligent search, retrieval and AI-assisted interaction with organizational knowledge.

2. Product Vision

Build the intelligence layer on top of enterprise documents. Instead of storing files, organizations store knowledge that can be searched, queried, summarized and automated.

3. Business Problem

Organizations spend significant time locating information spread across PDFs, Word documents, spreadsheets and presentations. Traditional keyword search is insufficient because it cannot understand context.

4. Objectives

• Centralized document management
• AI-based document understanding
• Semantic search
• Chat with documents
• Automation workflows
• Enterprise-ready architecture

5. System Architecture

Frontend → Backend (Go) → MinIO/PostgreSQL → gRPC → Python AI Service → Gemini Embeddings → Reverse gRPC Callback → Backend Persistence

6. Components

Backend: Gin, GORM, PostgreSQL, Redis, MinIO, gRPC Client & Callback Server.
AI Service: Loader Factory, Chunker, Embedding Service, Callback Client.
External: Gemini API, MinIO Object Storage.

7. Backend Responsibilities

Authentication, document upload, metadata persistence, storage management, orchestration, callback handling, API exposure and future search APIs.

8. AI Service Responsibilities

Download documents, detect loader, extract text, normalize content, generate chunks, generate embeddings, return processing results.

9. Document Lifecycle

Upload → Validate → Store → Create DB Record → Process → Chunk → Embed → Callback → Persist → Search Ready

10. Data Flow

HTTP uploads initiate processing. Backend persists metadata and calls the AI service via gRPC. The AI service processes the document and sends results back through reverse gRPC.

11. Database Responsibilities

Documents table stores metadata. Chunk and Embedding tables store processed content. Future pgvector indexes enable semantic retrieval.

12. Storage

Original files are stored in MinIO. Temporary processing occurs in isolated directories. Future versioning and lifecycle policies will be supported.

13. gRPC Communication

ProcessDocument is initiated by the backend. SaveProcessingResult is initiated by the AI service, eliminating polling and reducing latency.

14. Callback Architecture

Reverse callbacks notify the backend immediately after processing completes, enabling asynchronous workflows.

15. Error Handling

Retries for transient failures, structured logging, status tracking, idempotent processing and graceful recovery.

16. Security

Credential isolation, JWT authentication, role-based authorization (future), encrypted transport, input validation and secret management.

17. Logging & Monitoring

Structured logs, processing metrics, callback success rates, document throughput, latency dashboards and alerting.

18. Coding Standards

Layered architecture, dependency injection, interfaces, repository pattern, service separation, protobuf contracts and consistent logging.

19. Deployment

Independent deployment of Backend and AI Service using containers. Configuration through environment variables.

20. Disaster Recovery

Backups for PostgreSQL, MinIO replication, retry queues and document reprocessing.

21. Testing

Unit testing, integration testing, gRPC testing, API testing, loader validation and end-to-end processing verification.

22. Future Roadmap

Semantic search, pgvector, RAG chat, OCR, multilingual support, workflow automation, entity extraction, knowledge graph, collections, workspaces, collaboration, audit logs, analytics and public APIs.

Operational Checklist

Stage	Owner	Status
Upload	Backend	Validate request and upload to MinIO
Processing	AI Service	Load, chunk and embed document
Callback	AI Service	Send ProcessingResult
Persistence	Backend	Store chunks and embeddings
Completion	Backend	Update document status